OXFORD

INTERNATIONAL AQA EXAMINATIONS

Please write clearly in block capitals.

Centre number | | | | | | Candidate number | | | | |

Surname ________________________________

Forename(s) ________________________________

Candidate signature ________________________________

I declare this is my own work.

INTERNATIONAL GCSE
BIOLOGY
Paper 2

Thursday 1 November 2022   07:00 GMT   Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:
- a ruler with millimetre measurements
- a scientific calculator.

Instructions
- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information
- There are 90 marks available on this paper.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers.

Advice

In all calculations, show clearly how you work out your answer.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
TOTAL	

N O V 2 2 9 2 0 1 2 0 1

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9201/2

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ANSWER IN THE SPACES PROVIDED

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Answer **all** questions in the spaces provided.

0 1 This question is about an adaptation in birds.

0 1 . 1 Birds compete for food.

What else do birds compete for?

[1 mark]

Tick (✓) **one** box.

Light ☐

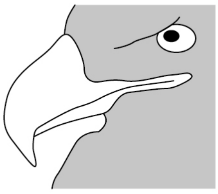
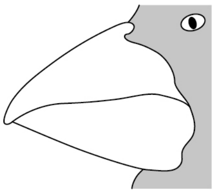
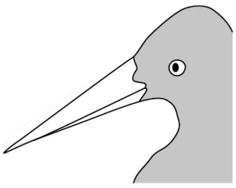
Oxygen ☐

Territory ☐

0 1 . 2 The shape of a bird's beak is adapted to the type of food it eats.

Draw **one** line from each beak shape to how the beak shape helps the bird to eat.

[3 marks]

Beak shape	How the beak shape helps the bird to eat
 Sharp and curved	Crunches and cracks seeds
 Short and thick	Filters small plants from the water
 Straight and pointed	Spears prey such as fish
	Tears apart prey such as mice

Turn over ►

A student used a model to investigate which beak shape would be best to pick up seeds.

The student used three different types of forceps to model a bird's beak and a plastic cup to model a bird's gut.

Figure 1 shows some of the equipment used in the investigation.

Figure 1

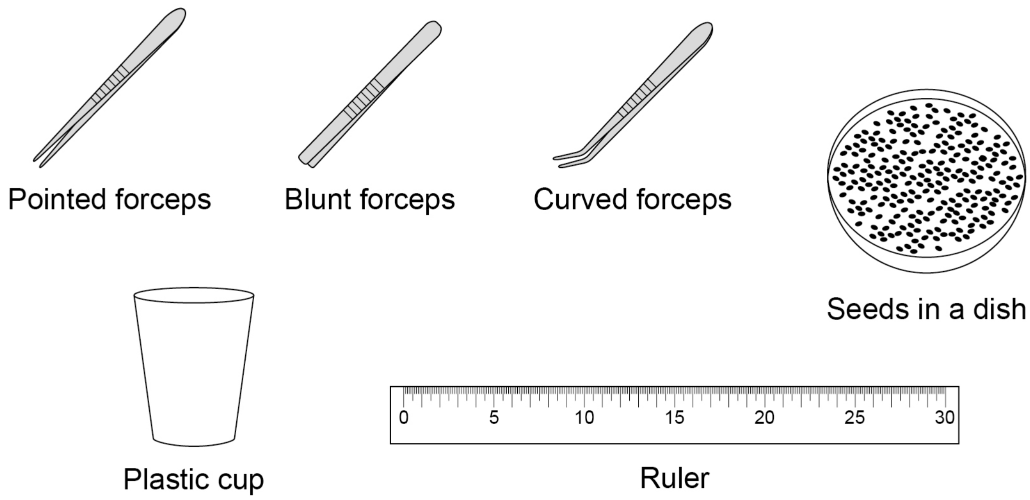


Table 1 shows steps in the method.

The steps are **not** in the correct order.

Table 1

A	Count how many seeds are in the cup after 60 seconds.
B	Place the cup 30 cm away from the dish of seeds.
C	Repeat using blunt forceps and then using curved forceps.
D	Use the pointed forceps to move one seed at a time from the dish into the cup.

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0 1 . 3 Write **one** letter in **each** of the three boxes to show the correct order for the student's method.

One of the boxes has been completed for you.

[2 marks]

Step 1	→	Step 2	→	Step 3	→	Step 4
						C

0 1 . 4 Name **one** other piece of equipment the student would need to complete the method given in **Table 1**.

Do **not** refer to the equipment in **Figure 1**.

[1 mark]

____

0 1 . 5 Which should be used to show the results?

[1 mark]

Tick (✓) **one** box.

Bar chart ☐

Line graph ☐

Scatter graph ☐

0 1 . 6 Some birds have a beak shape that allows them to eat a wide range of foods.

Give **one** advantage for a bird to be able to eat a wide range of foods.

[1 mark]

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9

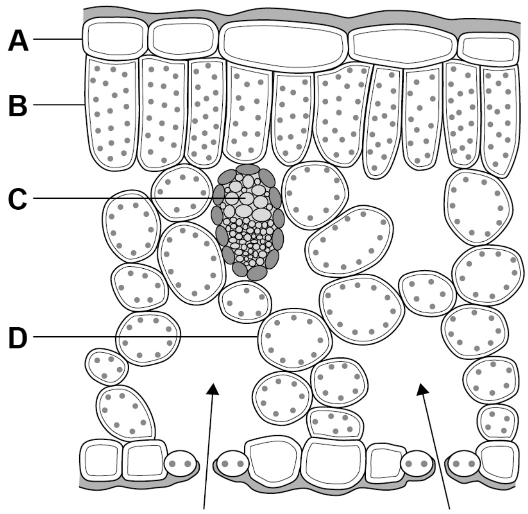
0 2 This question is about leaves. A leaf is a plant organ.

0 2 . 1 Name **one** other plant organ.

[1 mark]

Figure 2 shows the cross-section of part of a leaf.

Figure 2



0 2 . 2 The arrows on **Figure 2** represent the movement of a gas into the leaf during the day.

Which gas do the arrows represent?

[1 mark]

Tick (✓) **one** box.

Carbon dioxide ☐

Sulfur dioxide ☐

Water vapour ☐

0 2 . 3 Gases go into and out of the leaf through stomata.

Name the cells that control the size of the stomata.

[1 mark]

____

0 2 . 4 Which cell has the most chloroplasts?

[1 mark]

Tick (✓) **one** box.

A ☐

B ☐

C ☐

D ☐

0 2 . 5 Which cell is surrounded by many air spaces?

[1 mark]

Tick (✓) **one** box.

A ☐

B ☐

C ☐

D ☐

Question 2 continues on the next page

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0 2 . 6 Explain how the air spaces help in the movement of gases into and out of the **cells**.

[3 marks]

0 2 . 7 A leaf was viewed at a magnification of × 40

The thickness of the leaf in the image was 7 cm.

Calculate the real thickness of the leaf.

Give your answer in millimetres (mm).

Use the equation:

$$\text{Real size} = \frac{\text{size of image}}{\text{magnification}}$$

[3 marks]

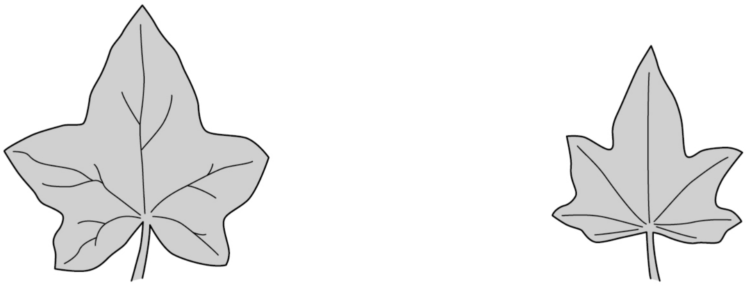
Real thickness of leaf = ____________________ mm

Do not write outside the box

0 2 . 8 Ivy is a plant that grows under trees in woodland.

Figure 3 shows two ivy leaves.

Figure 3



Leaf **A**
from the centre of a woodland

Leaf **B**
from the edge of a woodland

Explain why it is an advantage for leaf **A** to be larger than leaf **B**.

[3 marks]

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14

Turn over for the next question

0 3 The small intestine is an organ made of different types of tissue.

Each tissue has a particular function.

0 3 . 1 Draw **one** line from each type of tissue to the function of that type of tissue.

[2 marks]

Type of tissue	Function of tissue
Epithelial	Contracts to move contents through the small intestine
Glandular	Covers the outside of the small intestine
Muscular	Produces enzymes that are released in the small intestine

The enzyme amylase causes the breakdown of starch to sugars.

A student investigated the effect of pH on the activity of amylase.

This is the method used.

1. Place two drops of iodine solution into each well of a spotting tile.
2. Measure into a test tube:
   - 2 cm^3 of amylase solution
   - 2 cm^3 of starch solution
   - 1 cm^3 of pH 5 solution.
3. Mix the contents of the test tube.
4. After 30 seconds add a drop of the mixture from the test tube into one well of the spotting tile.
5. Repeat step 4 every 30 seconds until the iodine solution no longer turns a blue-black colour.
6. Repeat steps 1 to 5 using pH 7 solution and then pH 9 solution.

0 3 . 2 The risk assessment described the solutions used in the investigation as a hazard.

State **one** safety precaution that could reduce the risk of this hazard.

[1 mark]

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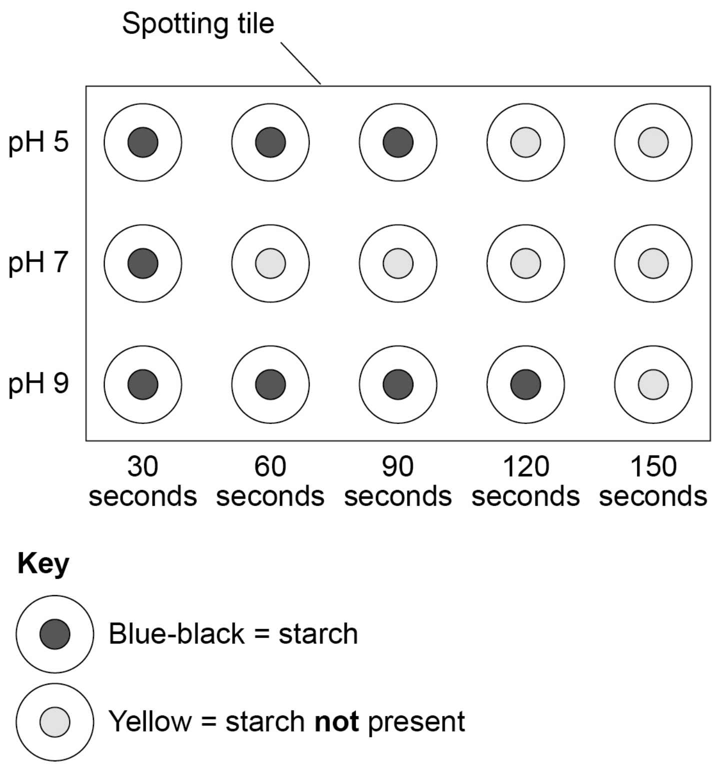
Figure 4 shows the results.

Figure 4



0 3 . 3 A control test was carried out at pH 7.

In the control test the method was repeated using 2 cm^3 of water instead of the amylase solution.

Describe the expected results for the control test.

[1 mark]

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Question 3 continues on the next page

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0 3 . 4 The student concluded that it took 60 seconds for **all** the starch to be broken down by amylase at pH 7.

In this test:

- the concentration of starch solution was 10 g/dm^3
- the volume of starch solution was 2 cm^3.

Calculate the mean **rate** of breakdown of starch at pH 7.

Give your answer:

- in g/s
- in standard form.

1 dm^3 = 1000 cm^3

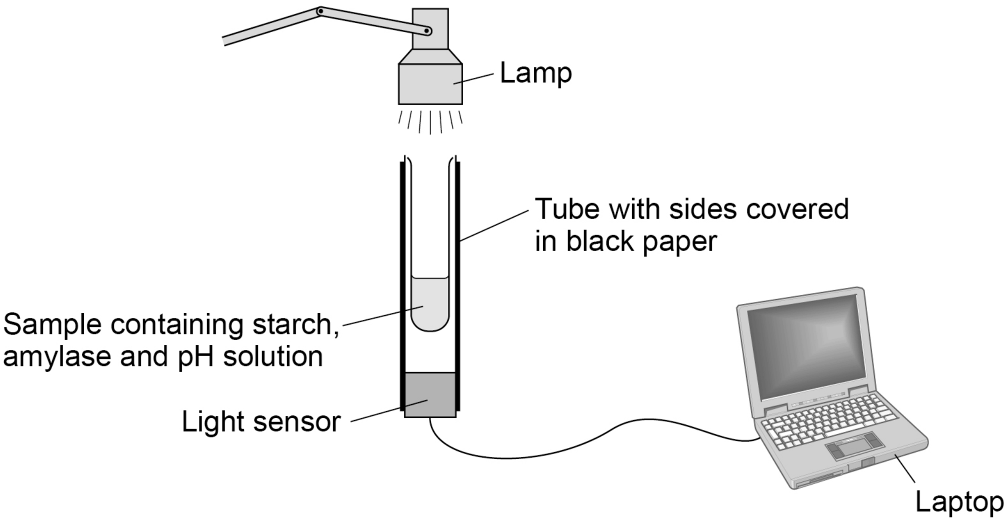
[5 marks]

Mean rate = ____________________ g/s

A light sensor can be used to investigate the effect of pH on the activity of amylase.

Figure 5 shows the equipment used.

Figure 5



The sample is cloudy at the start of the investigation due to insoluble starch.

The percentage of light that passes through the sample to the sensor is recorded every second.

The sample becomes clear as the starch is broken down and more light can pass through.

The results are displayed as a graph on the laptop.

Question 3 continues on the next page

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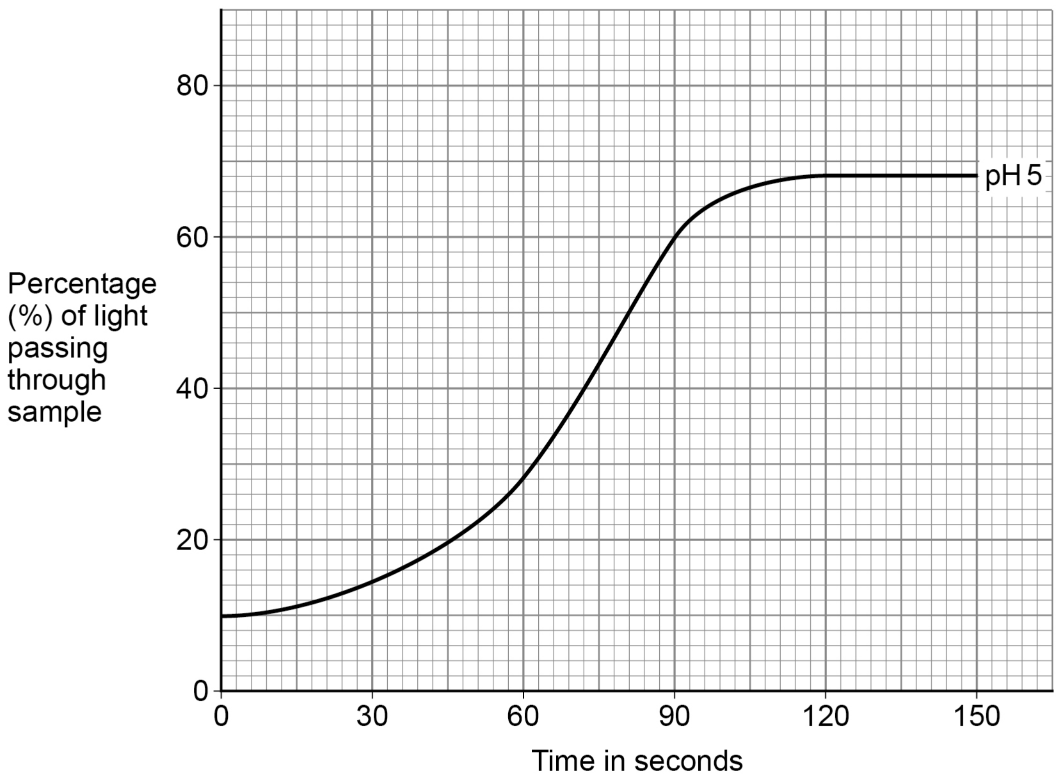
Figure 6 shows the results at pH 5.

Figure 6



0 3 . 5 Draw a line on **Figure 6** to show the expected results at pH 7.

Use **Figure 4** on page 11.

[2 marks]

Do not write outside the box

0 3 . 6 Suggest **three** reasons why the light sensor method is better than the spotting tile method.

[3 marks]

1 ____

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2 ____

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3 ____

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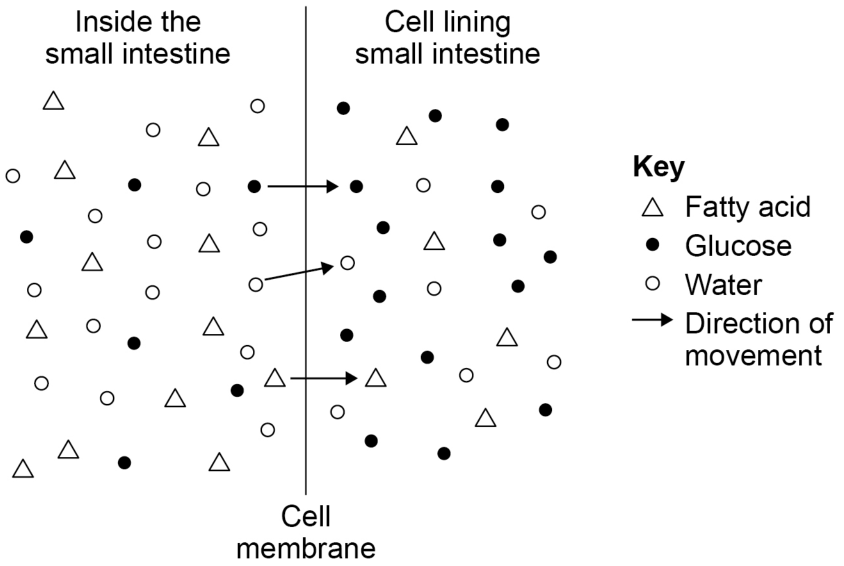
Question 3 continues on the next page

0 3 . 7 The small intestine is where most of the products of digestion are absorbed.
The products move from the inside of the small intestine into cells lining the small intestine.

Figure 7 represents the absorption of some substances in the small intestine.

Figure 7



Explain how fatty acids, glucose and water enter the cells lining the small intestine.

Use **Figure 7**.

[6 marks]

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20

Turn over for the next question

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0 4 Genetically modified (GM) plants can be produced by transferring genes from bacterial cells into plant cells.

0 4 . 1 A gene is a small section of DNA.

Describe the structure of DNA.

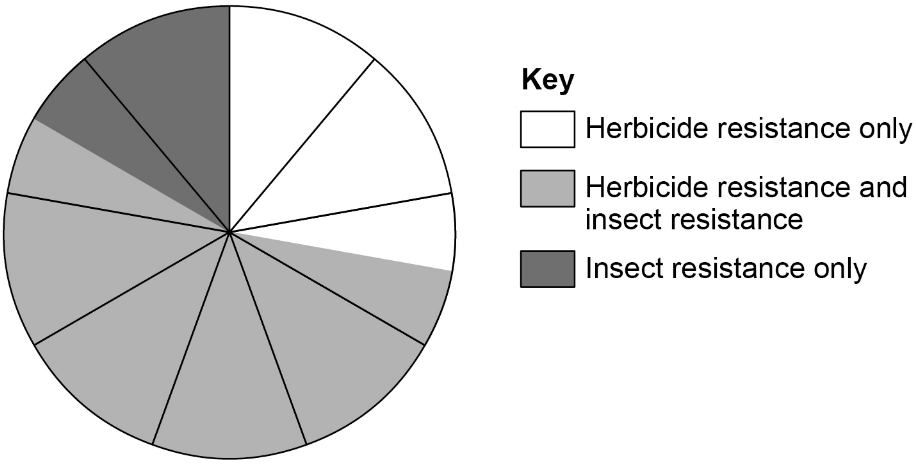
[2 marks]

Examples of genes transferred into plants include:

- a gene that gives herbicide resistance
- a gene that gives insect resistance.

Figure 8 shows how often these genes are used in GM plants in comparison to one another.

Figure 8



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0 4 . 2 Complete **Table 2**.

Use **Figure 8** to estimate the proportion of GM plants that have the genes for each type of resistance.

[2 marks]

Table 2

Type of resistance	Proportion
Herbicide resistance only	
Herbicide resistance and insect resistance	
Insect resistance only	

Question 4 continues on the next page

Do not write outside the box

0 4 . 3 A scientist investigated insect damage to the leaves of plants.

This is the method used.

1. Grow insect-resistant GM plants individually in pots in a greenhouse.
2. Add 4 insects of one species to each potted plant.
3. Measure the leaf area of each plant at the start of every month for 5 months.
4. Repeat steps 1 to 3 using the same insect species but with non-GM plants.

Table 3 shows the results.

Table 3

Month	Mean total leaf area per plant in cm^2	
	GM plants	Non-GM plants
0	158	152
1	98	92
2	80	64
3	72	44
4	74	32
5	80	27

The scientist concluded that the leaves of the GM plants were damaged less by insects compared to the leaves of the non-GM plants.

Evaluate the scientist's method and conclusion.

[4 marks]

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0 4 . 4 **Figure 9** shows some stages in the production of GM plants that are resistant to both a herbicide and to insect damage.

Figure 9

Both the gene for herbicide resistance and the gene for insect resistance are inserted into bacterial cells.

↓

These bacterial cells are mixed with plant cells to transfer the genes.

↓

The plant cells are grown in the presence of the herbicide.

↓

The plant cells that have taken up the genes develop into GM plants.

Explain why the plant cells are grown in the presence of the herbicide.

[2 marks]

10

Turn over for the next question

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0 5 The kidneys:

- filter the blood
- help control the water and ion content of the body.

0 5 . 1 Excess water and ions are excreted in urine.

Name **one** other substance excreted in the urine of a healthy person.

[1 mark]

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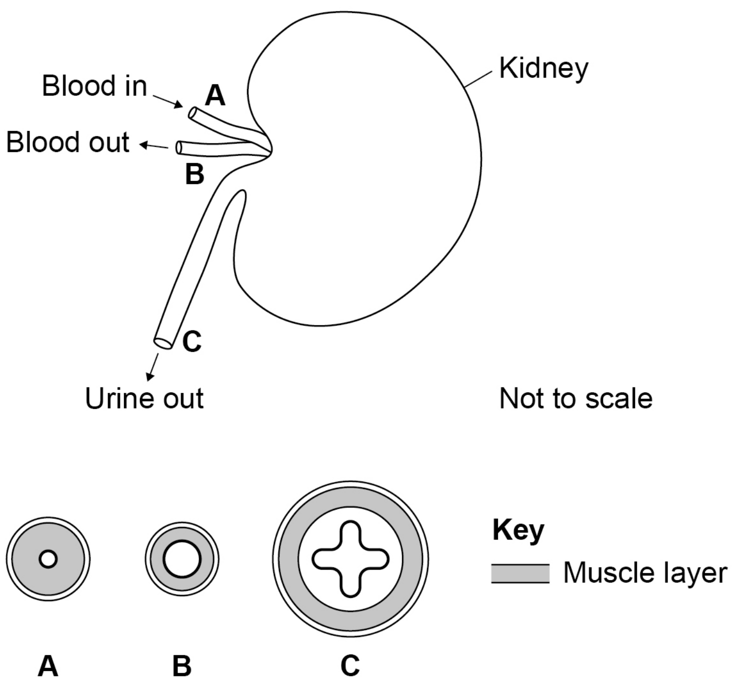
Figure 10 shows a human kidney and cross-sections of structures **A**, **B** and **C**.

Figure 10



Cross-sections of structures **A**, **B** and **C**

0 5 . 2 Which structure is an artery?

[1 mark]

Tick (✓) **one** box.

A ☐ **B** ☐ **C** ☐

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0 5 . 3 Give **two** reasons for your answer to Question **05.2**.

[2 marks]

1 ____

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2 ____

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An investigation was carried out to see how drinking pure water affects the production of urine.

This is the method used.

1. Ask a volunteer to sit for 30 minutes.
2. Collect and measure the volume of urine produced by the volunteer.
3. Ask the volunteer to drink 500 cm^3 of pure water.
4. Collect and measure the volume of urine 30 minutes after drinking the water.
5. Collect and measure the volume of urine 60 minutes after drinking the water.

0 5 . 4 The investigation was repeated with three more volunteers and similar results were obtained.

Give **one** advantage of other volunteers repeating the investigation.

[1 mark]

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____

0 5 . 5 Give **two** control variables which should be used when repeating the investigation.

[2 marks]

1 ____

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2 ____

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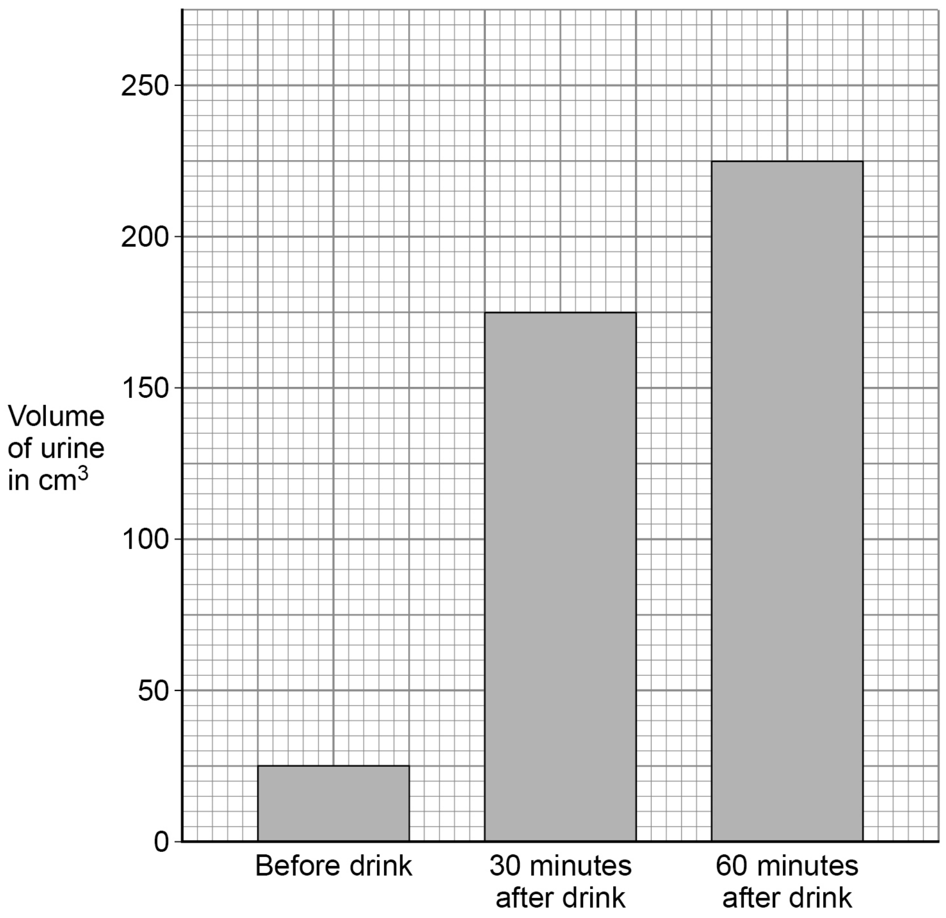
Figure 11 shows the results.

Figure 11



0 5 . 6 Explain why the volume of urine increased after the volunteer drank 500 cm^3 of pure water.

[4 marks]

Do not write outside the box

0 5 . 7 Explain what could happen to body cells if there was no control of the water content in the body after the drink.

[2 marks]

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13

Turn over for the next question

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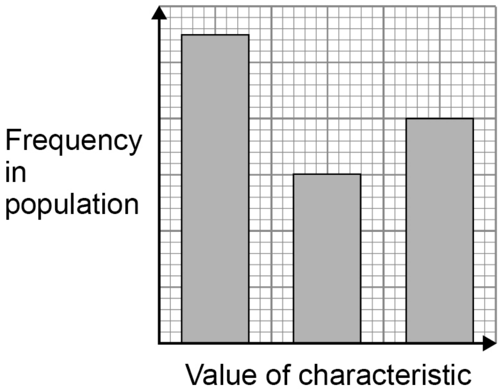
0 6 Individuals of the same species are not normally identical.

They have different characteristics from one another and show variation.

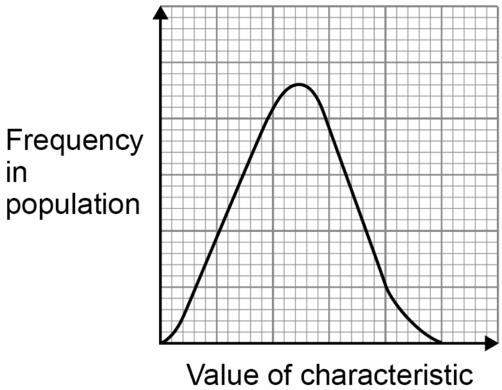
Figure 12 shows data about two types of variation.

Figure 12

Graph A



Graph B



0 6 . 1 How does **graph A** show that the data is discontinuous?

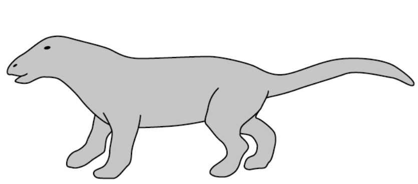
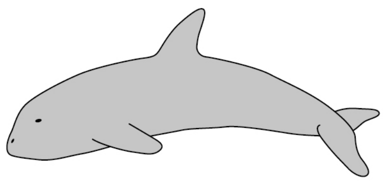
[1 mark]

In the 1800s, Darwin and Lamarck had different theories about how organisms evolved.

Lamarck thought that changes occurring in an organism during its lifetime could be inherited by its offspring.

Figure 13 shows an ancestor of the dolphin and a present-day dolphin.

Figure 13

Ancestor of the dolphin	**Present-day dolphin**
	
Lived on land.	Lives in water.
Had four legs.	Has flippers where the front legs used to be and a bone with no function where the back legs used to be.
Spent more and more time in the water getting food.	Eats fish and other seafood.

0 6 . 2 Explain how **Lamarck's** theory would account for the evolution of the water-living dolphin from the land-living ancestor.

[4 marks]

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Question 6 continues on the next page

Turn over ►

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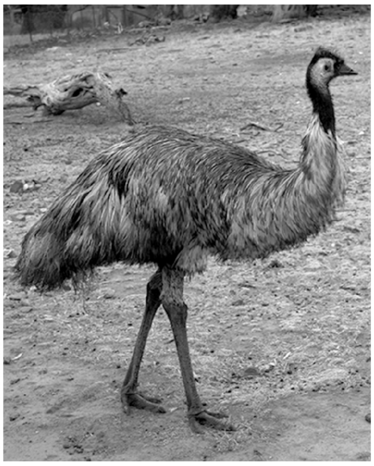
0 6 . 3 Give **one** reason why most scientists now accept Darwin's theory of evolution.

[1 mark]

Figure 14 shows an ostrich and an emu. These birds have many features in common even though they are two different species.

Figure 14



Ostrich
- lives in Africa
- flightless
- 1.8 to 2.7 metres tall
- long and powerful legs
- eats seeds and insects

Emu
- lives in Australia
- flightless
- 1.6 to 2.0 metres tall
- long and powerful legs
- eats seeds and insects

0 6 . 4 Suggest how scientists could prove that the ostrich and the emu are two different species.

[1 mark]

Do not write outside the box

0 6 . 5 Scientists believe ancestral birds of the ostrich and the emu flew long distances to new places.

These birds then independently evolved into different species of flightless birds.

Describe how the ostrich and the emu evolved to become two new species with many similar features.

Use **Figure 14**.

[5 marks]

12

Turn over for the next question

0 7 This question is about temperature regulation.

0 7 . 1 Describe the role of the skin in **monitoring** body temperature in humans.

[2 marks]

0 7 . 2 Explain why humans need more food when the environment becomes colder.

[3 marks]

0 7 . 3 Give **two** ways body temperature can be controlled during a period of intense exercise.

[2 marks]

1

2

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In humans the internal body temperature is regulated by homeostasis.

Butterflies do **not** have homeostatic processes.

Butterflies regulate their body temperature by changing their behaviour.

For example:

- they move into the sun to warm up
- they move into the shade to cool down.

0 7 . 4 Describe **one** advantage and **one** disadvantage of regulating body temperature using behavioural changes rather than using homeostatic processes.

[2 marks]

Advantage ______________________________

____

____

Disadvantage ______________________________

____

____

Question 7 continues on the next page

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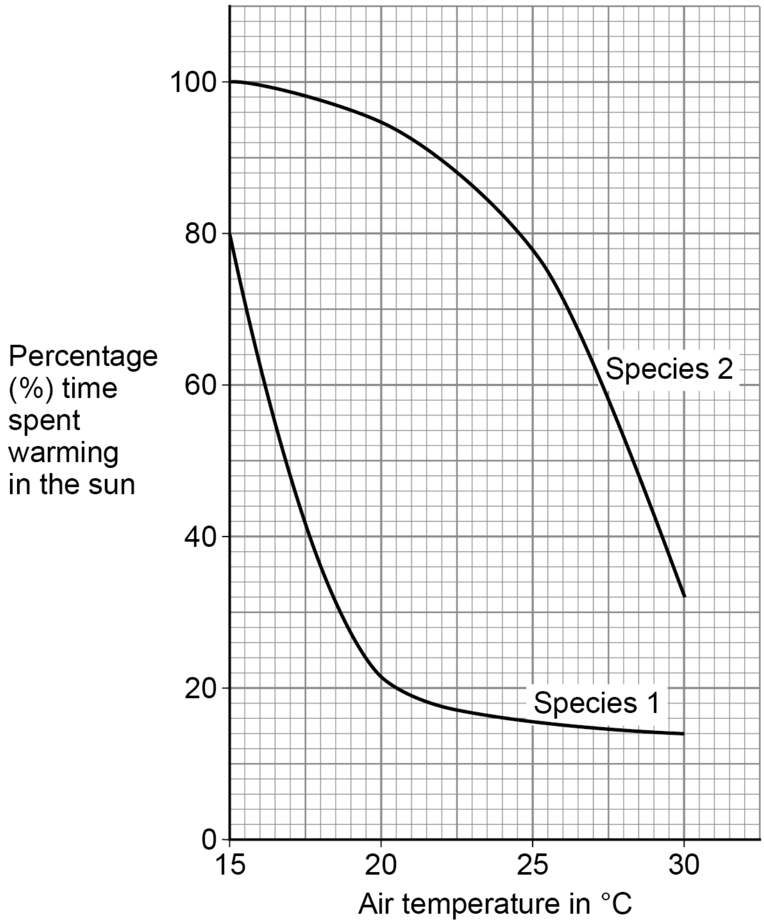
0 7 . 5 Two species of butterfly were monitored to find the effect of air temperature on the time they spent warming in the sun during the day.

Figure 15 shows the results.

Figure 15



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Compare the percentage time spent warming in the sun for Species **1** and Species **2** at different air temperatures.

Use data from **Figure 15**.

[3 marks]

12

END OF QUESTIONS

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